

Algebra and Discrete Mathematics (ADM)

Tutorial 2 Linear systems and Gauss–Jordan elimination

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Solving a system of linear equations

$$\begin{aligned}2x + y &= 5 \\ x - y &= -2\end{aligned}$$

By substitution, the second equation implies $x = -2 + y$. Substituting this into the first equation gives

$$2(-2 + y) + y = 5 \implies 3y = 9 \implies y = 3 \implies x = 1$$

By elimination, adding the two equations gives

$$3x = 3 \implies x = 1$$

$$1 - y = -2 \implies y = 3$$

Solving a system of linear equations

A more complicated system

$$-x + 4y + z = -5$$

$$2x + 2y + z = 3$$

$$x - 2y - z = 3$$

Convert the system to matrix form:

$$\begin{pmatrix} -1 & 4 & 1 \\ 2 & 2 & 1 \\ 1 & -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \\ 3 \end{pmatrix}$$

Augmented matrix

$$\left(\begin{array}{ccc|c} -1 & 4 & 1 & -5 \\ 2 & 2 & 1 & 3 \\ 1 & -2 & -1 & 3 \end{array} \right)$$

Solving a system of linear equations

$$\left(\begin{array}{ccc|c} -1 & 4 & 1 & -5 \\ 2 & 2 & 1 & 3 \\ 1 & -2 & -1 & 3 \end{array} \right)$$

Elementary row operations for Gauss--Jordan elimination

- Multiply a row by a nonzero constant
- Interchange two rows
- Add a constant times one row to another

Solving a system of linear equations

$$\left(\begin{array}{ccc|c} -1 & 4 & 1 & -5 \\ 2 & 2 & 1 & 3 \\ 1 & -2 & -1 & 3 \end{array}\right) \xrightarrow{R_1 \rightarrow -1R_1} \left(\begin{array}{ccc|c} 1 & -4 & -1 & 5 \\ 2 & 2 & 1 & 3 \\ 1 & -2 & -1 & 3 \end{array}\right)$$

$$\xrightarrow[\begin{array}{l} R_3 \rightarrow R_3 - R_1 \\ R_2 \rightarrow R_2 - 2R_1 \end{array}]{R_2 \rightarrow \frac{1}{10}R_2} \left(\begin{array}{ccc|c} 1 & -4 & -1 & 5 \\ 0 & 10 & 3 & -7 \\ 0 & 2 & 0 & -2 \end{array}\right) \xrightarrow{R_2 \rightarrow \frac{1}{10}R_2} \left(\begin{array}{ccc|c} 1 & -4 & -1 & 5 \\ 0 & 1 & \frac{3}{10} & -\frac{7}{10} \\ 0 & 2 & 0 & -2 \end{array}\right)$$

$$\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left(\begin{array}{ccc|c} 1 & -4 & -1 & 5 \\ 0 & 1 & \frac{3}{10} & -\frac{7}{10} \\ 0 & 0 & -\frac{3}{5} & -\frac{3}{5} \end{array}\right) \xrightarrow{R_3 \rightarrow -\frac{5}{3}R_3} \left(\begin{array}{ccc|c} 1 & -4 & -1 & 5 \\ 0 & 1 & \frac{3}{10} & -\frac{7}{10} \\ 0 & 0 & 1 & 1 \end{array}\right)$$

We have reached row echelon form

Solving a system of linear equations

Backward:

$$\left(\begin{array}{ccc|c} 1 & -4 & -1 & 5 \\ 0 & 1 & \frac{3}{10} & -\frac{7}{10} \\ 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow[R_1 \rightarrow R_1 + R_3]{R_2 \rightarrow R_2 - \frac{3}{10} R_3} \left(\begin{array}{ccc|c} 1 & -4 & 0 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + 4R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

This is the reduced row echelon form.

Solving system of linear equations

$$-x + 4y + z = -5$$

$$2x + 2y + z = 3$$

$$x - 2y - z = 3$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

The solution is given by

$$x = 2, \quad y = -1, \quad z = 1$$

Gauss–Jordan elimination

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & -1 & 0 \\ 2 & 2 & 1 & 5 \end{array}\right)$$

The leftmost nonzero column is the first column, and it already contains a leading 1.

$$\begin{array}{c} R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array} \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -2 & -2 & -4 \\ 0 & -2 & -1 & -3 \end{array}\right) \begin{array}{c} R_4 \rightarrow R_4 + 2R_2 \\ R_3 \rightarrow R_3 + 2R_2 \end{array} \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 3 \end{array}\right)$$

$$\xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & 3 \end{array}\right) \xrightarrow{R_4 \rightarrow R_4 - 3R_3} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

We have reached row echelon form with Gaussian elimination

Gauss–Jordan elimination

Backward

$$\begin{pmatrix} 1 & 2 & 1 & | & 4 \\ 0 & 1 & 2 & | & 3 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & -3 & | & -2 \\ 0 & 1 & 2 & | & 3 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$
$$\xrightarrow[\begin{matrix} R_2 \rightarrow R_2 - 2R_3 \\ R_1 \rightarrow R_1 + 3R_3 \end{matrix}]{\begin{matrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{matrix}} \begin{pmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

This is the reduced row echelon form.

Gauss–Jordan elimination

$$\begin{pmatrix} 1 & 2 & -1 & \left| & \alpha \\ 2 & 3 & -2 & \left| & \beta \\ -1 & -1 & 1 & \left| & \gamma \end{pmatrix} \xrightarrow[R_2 \rightarrow R_2 - 2R_1]{R_3 \rightarrow R_3 + R_1} \begin{pmatrix} 1 & 2 & -1 & \left| & \alpha \\ 0 & -1 & 0 & \left| & \beta - 2\alpha \\ 0 & 1 & 0 & \left| & \gamma + \alpha \end{pmatrix} \xrightarrow{R_2 \rightarrow -R_2} \right. \\ \left. \begin{pmatrix} 1 & 2 & -1 & \left| & \alpha \\ 0 & 1 & 0 & \left| & 2\alpha - \beta \\ 0 & 1 & 0 & \left| & \gamma + \alpha \end{pmatrix} \xrightarrow[R_1 \rightarrow R_1 - 2R_2]{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 & \left| & -3\alpha + 2\beta \\ 0 & 1 & 0 & \left| & 2\alpha - \beta \\ 0 & 0 & 0 & \left| & \gamma - \alpha + \beta \end{pmatrix} \right. \right.$$

There are two cases:

- If $\gamma - \alpha + \beta \neq 0$, the system is inconsistent and has no solution.
- If $\gamma - \alpha + \beta = 0$, the last row is a zero row and the system is consistent.

In the consistent case, z is a free variable. Letting $z = t$, where $t \in \mathbb{R}$, we obtain

$$x = -3\alpha + 2\beta + t, \quad y = 2\alpha - \beta, \quad z = t.$$

Hence the solution set is

$$\{ (-3\alpha + 2\beta + t, 2\alpha - \beta, t) \mid t \in \mathbb{R} \}.$$